

Project Description

This course project will give students the opportunity to learn about cool methods in machine learning; do research; and get your hands dirty with data. Please **work in groups of 3 students on the projects**.

Something **new** should be **learned** as a result of your project. Your project may innovate some new ML methodology, or make an improvement to an existing method. It is also ok to learn how to apply existing ML models in a new application area. However, it is not ok to swap out one datafile for another in an existing script, or to produce a study whose main contribution is making a call to a “.fit()” function from an existing library. (This project will be judged against a semester’s worth of effort.) If your project is application based, it must: compare across multiple methods, hyperparameters, and ways of pre-processing data; make use of multiple datasets; interpret results and models; list best practices for future researchers in the area with empirical (and possibly theoretical) backing.

Deliverables

The main **deliverables** are the following:

1. A project write-up, 8 pages [NeurIPS](#) format. (Note this requires using LaTeX, the standard markup language for scientific and technical manuscripts.)
2. Open source repository with executable code for methods developed.
3. Spotlight Presentation.

Project Midway Report

Please write a 4 page midway report as a pdf with the [NeurIPS style files](#) with the following:

- Introduction Section
 - Introduce and motivate the problem and methods.
- Related Work
 - Summarize your literature review and related approaches.
- Methods
 - Expound on your methodology; be precise and mathematical.
- Preliminary Results
 - Summarize the current quantitative and qualitative results you have obtained.
- Future Plans
 - Provide an outline (and division of labor) with estimated dates of completion for the rest of the project.

The 4 pages do not include references (have as many as needed). Please do not use frivolous whitespace and figures; i.e. I expect that each of the sections above is well developed with text. Do not use bullets; the sections (especially the Intro and Related Work sections) should serve

as a draft for your final report. Submit your proposal as a pdf file. Include all names in the proposal text and file name.